

Machine Learning Fundamentals

Chapter 1: Introduction to Machine Learning

Machine learning is a branch of artificial intelligence in which computer systems learn patterns from data and use those patterns to make predictions or decisions.

A typical machine learning workflow begins with collecting data, cleaning it, selecting useful features, training a model, evaluating the model, and finally deploying it for inference.

Supervised learning uses labeled examples. Classification predicts categories such as spam or not spam, while regression predicts numerical values such as house prices.

Key Points

- Training data contains examples used to learn patterns.
- Validation data helps tune model choices.
- Test data measures performance on unseen examples.

Chapter 2: Model Training and Evaluation

Training is the process of adjusting a model's parameters so that its predictions become closer to the expected outputs. A loss function measures prediction error, and an optimizer changes model parameters to reduce that loss.

Gradient descent is a common optimization method. It calculates the direction in which the loss increases and updates parameters in the opposite direction. The learning rate controls the size of each update.

Overfitting occurs when a model learns the training examples too closely and performs poorly on new data. Underfitting occurs when a model is too simple to capture useful patterns.

Common evaluation metrics include accuracy, precision, recall, F1 score, and mean squared error. The appropriate metric depends on the problem and the consequences of different errors.

Key Points

- Use a validation set for model selection.
- Tune the learning rate and other hyperparameters carefully.
- Use regularization or more data to reduce overfitting.
- Choose evaluation metrics based on the task.

Chapter 3: Neural Networks

A neural network consists of layers of interconnected computational units. Each unit combines inputs with learned weights, adds a bias, and applies an activation function.

During forward propagation, information moves from the input layer through hidden layers to the output layer. During backpropagation, the model calculates gradients of the loss with respect to its parameters.

Deep learning refers to neural networks with multiple layers. These models can learn useful representations from large datasets and are widely used for computer vision, speech, and natural language processing.

Key Points

- Weights and biases are learned parameters.
- Activation functions introduce non-linearity.
- Backpropagation computes gradients used during optimization.